

Student-Created MCQs for SG10

1. A 44-year-old woman with a history of hypertension presents with a sudden onset of dizziness, difficulty swallowing, and a hoarse voice. She reports slurred speech and noticed a loss of pain and temperature over the left side of her face. She stated that she can't feel pain in her right leg and arm. Physical exam shows ataxia on the left side and mild ptosis on the same side. Which of the following best characterizes the clinical syndrome associated with these findings?

- A. Medial Pontine Syndrome
- B. Medial Medulla Syndrome
- C. Lateral Medullary Syndrome
- D. Lateral Pontine Syndrome

2. A 60-year-old man with a history of poorly controlled hypertension presents with sudden-onset facial weakness, dizziness, and difficulty walking. Neurologic exam shows paralysis of the left side of the face, left-sided hearing loss, and ataxia on the left. He also has decreased pain and temperature sensation over the right arm and leg. MRI confirms a vascular lesion affecting the left lateral aspect of the caudal pons. Which of the following additional findings is most likely to be present in this patient?

- A. Dysphagia and hoarseness
- B. Tongue deviation to the left on protrusion
- C. Decreased pain and temperature sensation on the left side of the face
- D. Hyperreflexia in the left limbs
- E. Complete loss of proprioception on the right side of the body

3. A 65-year-old man is brought to the emergency room after suddenly developing dizziness, nausea, and an unstable gait. He reports his voice has become hoarse, and he has trouble swallowing. On examination, he has decreased pain & temperature sensation on the left side of his face and right side of his body. He also has left-sided ptosis and miosis, and there is ataxia of the left limbs and nystagmus. Which of the following arteries is most likely occluded in this patient?

- A. Middle cerebral artery
- B. Anterior spinal artery
- C. Posterior inferior cerebellar artery
- D. Posterior cerebral artery
- E. Anterior inferior cerebellar artery

4. A 65-year-old man presents to the emergency department with a 2-hour history of persistent hiccups. He also reports dizziness and difficulty swallowing. Neurologic examination reveals decreased pain and temperature sensation on the left side of his face and the right side of his body. Gag reflex is diminished on the left. Which of the following arteries most likely affected?

- A. Anterior spinal artery
- B. Anterior inferior cerebellar artery (AICA)
- C. Posterior cerebral artery (PCA)
- D. Posterior inferior cerebellar artery (PICA)
- E. Basilar artery

5. A 64-year-old woman comes to the emergency department with sudden onset of dizziness, nausea, hoarseness, and difficulty swallowing. On neurological exam, she has decreased pain and temperature sensation over the right side of her body. She also has left-sided Horner syndrome and ataxia. Her palate elevates asymmetrically, with the uvula deviated to the right. The patient has a hoarse voice and an absent gag reflex on the left. Which of the following structures is most likely affected in this patient?

- A. Medial lemniscus
- B. Hypoglossal nucleus
- C. Spinothalamic tract
- D. Oculomotor nerve
- E. Internal capsule

6. A 64-year-old man presents to the emergency department with sudden onset of facial droop and numbness. He also reports dizziness and difficulty walking. On physical examination, he has decreased pain and temperature sensations on the right side of his face, along with drooping of the right side of his mouth and an inability to close his right eye. The sensation of touch on the face is intact. There is also decreased pain and temperature sensation on the left upper and lower extremities. Which of the following best explains the patient's right-sided facial sensory loss?

- A. Damage to the right principal sensory nucleus of CN V
- B. Damage to the left spinothalamic tract
- C. Damage to the right spinal trigeminal tract
- D. Damage to the left trigeminothalamic tract
- E. Damage to the right facial motor nucleus

7. A 70-year-old man is brought to the emergency department after developing sudden weakness of the left side of his body. His wife reports that he also began having difficulty moving his eyes, especially when trying to look to the right. On examination, the patient is alert and oriented. He is unable to abduct his right eye, and both eyes fail to move past the midline when attempting to look toward the right. His right eye remains medially deviated at rest. Strength is 2/5 in the left upper and lower extremities. Vibration and proprioception are decreased on the left side of the body. Pain and temperature sensations are intact. MRI reveals an acute infarct involving the right medial pons. Which of the following arteries is most likely occluded?

- A. Anterior inferior cerebellar artery (AICA)
- B. Posterior inferior cerebellar artery (PICA)
- C. Posterior cerebral artery (PCA)
- D. Paramedian branches of the basilar artery
- E. Superior cerebellar artery

8. A 58-year-old woman is brought to the emergency department after developing sudden-onset dizziness, nausea, and difficulty walking. She also complains of decreased hearing in her left ear. On physical examination, she has left-sided facial weakness and decreased pain and temperature sensation on the right side of her body. There is also loss of pain and temperature sensation on the left side of her face. She exhibits gait instability and falls toward the left when attempting to stand with her eyes open. Audiometry confirms sensorineural hearing loss on the left. Which of the following arteries is most likely occluded?

- A. Right posterior inferior cerebellar artery (PICA)
- B. Left posterior inferior cerebellar artery (PICA)
- C. Left anterior inferior cerebellar artery (AICA)
- D. Right anterior inferior cerebellar artery (AICA)
- E. Basilar artery
- F. Right Vertebral artery
- G. Left vertebral artery
- H. Right superior cerebellar artery
- I. Left superior cerebellar artery

9. A 65-year-old man is brought to the emergency department after suddenly developing dizziness, difficulty swallowing, and hoarseness. On examination, he has decreased pain and temperature sensation on the left side of his face and the right side of his body. His uvula deviates to the right, and he has ataxia on the left side. A CT angiogram reveals an infarct in the territory of a major artery supplying the lateral medulla. Which of the following additional findings is most likely to be present, and Which artery is most likely involved?

- A. Left-sided hemiparesis; anterior spinal artery
- B. Left-sided Horner syndrome; posterior inferior cerebellar artery
- C. Bilateral leg weakness and sensory loss; anterior cerebral artery
- D. Ophthalmoplegia and ataxia; basilar artery
- E. Left-sided facial droop and hearing loss; posterior inferior cerebellar artery

10. A 45-year-old man comes to the physician because of a sudden loss of touch, proprioception, and vibration on the left side of the body. MRI shows occlusion of the right anterior spinal artery. Which of the following additional findings is most likely present?

- A. Loss of pain and temperature sensation on the left side of the body
- B. Loss of pain and temperature sensation on the right side of the face
- C. Tongue deviation toward the right on protrusion
- D. Ptosis and miosis on the right side
- E. Inability to clench the jaw

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